



Aluminum-Core / Metal-Core PCBs

MCPCB — when FR-4 can't sink the heat

The standard for LED, motor drive, and high-power surface-mount

Audience: Power + LED engineers

Construction • Thermal • Single-layer limits • Use cases • Cost



What is MCPCB

- ▶ MCPCB = Metal-Core Printed Circuit Board
- ▶ Base substrate is METAL (typically aluminium, sometimes copper)
- ▶ Thin thermally-conductive dielectric layer (75-200 μm) bonds Cu trace layer
- ▶ Heat from components $\rightarrow$ Cu trace $\rightarrow$ dielectric $\rightarrow$ metal base $\rightarrow$ enclosure or heatsink
- ▶ Thermal conductivity: 1-3 W/mK for dielectric (vs 0.3 W/mK for FR-4)
- ▶ 10 $\times$ better heat dissipation than standard FR-4
- ▶ Used in: LED lighting, motor drivers, automotive lighting, power supplies

MCPCB Construction

- ▶ 3-layer stack: Cu trace + dielectric + metal base
- ▶ Cu thickness: 1-3 oz typical (35-105 μm)
- ▶ Dielectric: 75-200 μm thermal epoxy (filled with ceramic for conductivity)
- ▶ Metal base: aluminium 1100/5052 (most common), or copper for highest thermal
- ▶ Total thickness: 1.0-2.0 mm typical (similar to FR-4)
- ▶ Single-layer only — you can't put a trace ON TOP of the metal core's other side
- ▶ For multi-layer: composite of FR-4 + metal-core OR special multilayer MCPCB (rare, costly)

Aluminum vs Copper Core

Aluminium core

- ▶ Most common; 70% of MCPCBs
- ▶ Thermal conductivity: 230 W/mK (aluminium)
- ▶ Cheaper than copper
- ▶ Lighter weight
- ▶ Good for LED lighting bars
- ▶ Easy to machine + drill

Copper core

- ▶ Used for highest thermal performance
- ▶ Thermal conductivity: 400 W/mK
- ▶ ~2× cost vs aluminium
- ▶ Heavier
- ▶ Used in EV power modules
- ▶ Better for vibration resistance

Typical Applications

LED lighting

Strip lights, street lights
LED dies dissipate 5-10 W each

Automotive lighting

Headlights, brake lights
LED matrix with thermal

Motor drives

BLDC, stepper drivers
MOSFETs + heat

Power LEDs

Spot, flood, grow lights
Multiple high-power LEDs

Solar inverters

Microinverters mounted to panels
Thermal demands

Welding

Inverter-style welders
High-current power stages

Thermal Analysis Example

- ▶ Scenario: 10 W LED on a 30×30 mm MCPCB
- ▶ Dielectric thermal resistance: 0.5 K/W (75 μm at 2 W/mK over 9 cm²)
- ▶ Aluminium base resistance to ambient: 5 K/W (with heatsink)
- ▶ Total R_{th}: 5.5 K/W (LED junction to ambient)
- ▶ Junction temp = 25 °C + (10 W × 5.5) = 80 °C — well within LED T_{jmax} (125 °C)
- ▶ Same setup on FR-4: dielectric R_{th} = 30+ K/W → T_j = 325 °C → LED dies in hours
- ▶ MCPCB makes high-power LED designs possible

MCPCB Limitations

- ▶ Single-layer only (almost always) — limits design complexity
- ▶ No plated through-holes — metal base prevents PTH
- ▶ Drill holes are blind only (don't go through metal) OR through-hole with insulating sleeve
- ▶ Vias must be carefully designed (insulating gap around drilled hole)
- ▶ Cu weight typically 1 oz; heavier requires specialty fab
- ▶ Higher dielectric loss than FR-4 (not suitable for high-speed RF)
- ▶ Bend resistance: limited (don't bend an MCPCB without supporting structure)

Design Workflow

- ▶ Same EDA tool (KiCad / Altium) as FR-4 — stackup just becomes single-layer Cu + dielectric + metal
- ▶ Layout: all components on top side; ground on bottom (metal base often floating or chassis ground)
- ▶ Vias: blind to inner dielectric only (no through-hole)
- ▶ Power + ground routing: large pours for thermal spreading
- ▶ Cooling: design with assembly to heatsink in mind
- ▶ Hole spec: metal base needs separate hole spec from copper layer
- ▶ Soldering: standard SMT reflow works on MCPCB

MCPCB Fabrication

- ▶ Process: laminate dielectric + Cu foil onto pre-cut aluminium plate
- ▶ Etch Cu layer in standard PCB etcher
- ▶ Apply solder mask + silkscreen (similar to FR-4)
- ▶ CNC route board outline (aluminium is easy to machine)
- ▶ Drill any through-holes carefully (no copper plating)
- ▶ Cost: 1.5-3× standard FR-4 PCB
- ▶ Lead time: 5-15 days (specialty fabs only)
- ▶ Suppliers: Bergquist, Henkel, Henkel Bergquist (now combined)

Soldering MCPCB

- ▶ Reflow soldering works — same SMT process
- ▶ Preheat is critical: MCPCB has high thermal mass (metal base absorbs heat)
- ▶ Reflow profile: extend soak phase to thoroughly heat the board
- ▶ Peak temperature: same as FR-4 (245 °C for SAC305)
- ▶ Cool-down can be slower (metal stores heat)
- ▶ Hand-soldering: needs higher-power iron (40-60 W minimum, 80 W better)
- ▶ Components on the BOTTOM of MCPCB are NOT typical — top-side assembly only

Mechanical Assembly

- ▶ MCPCB usually mounts directly to a heatsink or enclosure
- ▶ Bottom of metal core: bare aluminium — thermal contact surface
- ▶ Thermal interface material (TIM) between MCPCB and heatsink:
 - Thermal paste (best thermal performance)
 - Thermal pad / SilPad (easier assembly, slightly less thermal performance)
 - Phase-change pad (premium, conforms when heated)
- ▶ Mounting: screws through edge holes (insulated from circuit)
- ▶ Galvanic isolation: metal base usually NOT electrically connected to circuit

Common Mistakes

- ▶ Specifying multi-layer MCPCB without confirming fab capability (rare)
- ▶ Trying to drill through-holes — metal base prevents standard PTH
- ▶ Treating MCPCB like FR-4 — forgetting blind via only
- ▶ Insufficient thermal interface to heatsink → most of MCPCB's benefit lost
- ▶ Hand-soldering with low-power iron → cold joints
- ▶ Forgetting reflow preheat → solder doesn't melt evenly
- ▶ Mounting screws shorting to circuit (no insulating washer)
- ▶ MCPCB for low-power LED → cost overhead with no benefit

Key Takeaways

Key takeaways from this lecture

MCPCB = thermal champion

10× better than FR-4.

Single-layer only

Major limitation.

Aluminium most common

Copper for premium thermal.

TIM to heatsink critical

Don't lose the thermal benefit.

Cost 1.5-3× FR-4

Worth it for power LEDs.

Reflow with extra preheat

Metal mass = slow heating.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ Henkel Bergquist — MCPCB technical documentation
- ▶ Cree / Osram LED app notes (use MCPCB)
- ▶ Texas Instruments LED driver reference designs
- ▶ Phil's Lab — MCPCB walkthrough
- ▶ Indian fabs offering MCPCB — check before specifying
- ▶ Aluminium grades: 1100, 5052 — strength + thermal trade-offs

Thank you • Build something this week.